Smart Music Intelligence System for Mood Prediction and Personalized Recommendation

Aditi Singh, Samridhi, Adhya Singh Chauhan, Khushi Bisht
School of Computer Science (SoCS)
UPES, Dehradun, India

***Abstract*—With the rapid growth of digital music platforms, users are exposed to an overwhelming number of songs, making music discovery a challenging task. This paper presents a Smart Music Intelligence System that integrates mood prediction, popularity estimation, and personalized recommendation. The system utilizes audio features such as energy, danceability, tempo, valence, and loudness to model user preferences.**

A Random Forest classifier is employed for mood prediction, while a Random Forest regressor estimates song popularity. Additionally, a K-Nearest Neighbors model using cosine similarity is implemented to recommend similar songs. The system is deployed using Streamlit, providing an interactive interface where users can explore and refine their music choices.

Experimental results demonstrate that combining predictive modeling with recommendation techniques enhances user engagement and improves the overall music discovery experience. The system also highlights how feature-based modeling can effectively capture emotional aspects of music.

***Index Terms*—Machine Learning, Music Recommendation, Random Forest, KNN, Mood Prediction**

I. Introduction

Music has become an essential part of everyday life, with streaming platforms providing access to millions of tracks. While this abundance offers variety, it also introduces complexity in identifying relevant content. Traditional recommendation systems rely heavily on user history or collaborative filtering, often ignoring emotional context.

Human emotions play a crucial role in music selection. People tend to choose music based on their mood, environment, or activity. Therefore, incorporating mood prediction into recommendation systems can significantly improve user satisfaction.

This paper proposes a system that combines machine learning models with user interaction to deliver personalized recommendations. By integrating mood classification, popularity prediction, and similarity-based recommendation, the system provides a more comprehensive and engaging user experience.

II. Related Work

Music recommendation systems have evolved over time from basic filtering techniques to advanced machine learning approaches. Early systems primarily relied on collaborative filtering, which uses user behavior patterns to suggest content.

Recent approaches focus on content-based filtering, where song features such as tempo, rhythm, and energy are analyzed. Machine learning models, including decision trees and clustering algorithms, have been widely used for classification and recommendation tasks.

Mood-based recommendation systems have gained attention due to their ability to align music with user emotions. However, many existing systems lack integration between prediction and recommendation. This work addresses that gap by combining multiple techniques into a unified framework.

III. Methodology

A. Feature Selection and Scaling

The system uses five key features: energy, danceability, tempo, valence, and loudness. These features are normalized using MinMax scaling:

$$X' = \frac{X - X_{min}}{X_{max} - X_{min}}$$

This ensures that all features contribute equally during model training.

B. Mood Classification

A Random Forest classifier is used to predict mood categories such as Energetic, Calm, Sad, Party, and Happy:

$$\hat{y} = \text{mode}(T_1(x), ..., T_n(x))$$

C. Popularity Prediction

A Random Forest regressor estimates song popularity based on input features:

$$\hat{y} = \frac{1}{n} \sum T_i(x)$$

D. Recommendation System

A KNN model is used for recommendation using cosine similarity:

$$\text{Similarity} = \frac{A \cdot B}{||A||||B||}$$

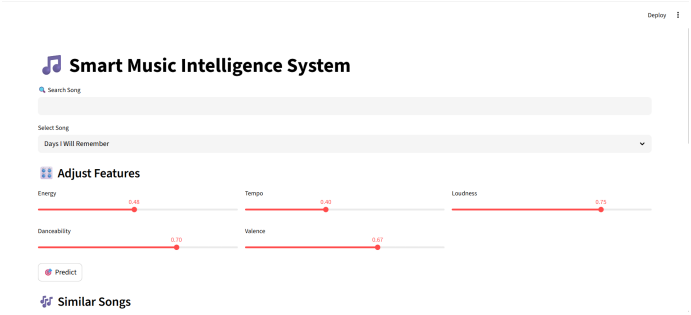

Fig. 1. User Interface Before Prediction Showing Feature Controls

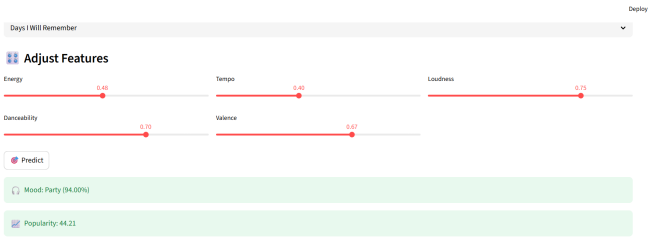

Fig. 2. Prediction Output Showing Mood and Popularity

E. System Implementation

The system is implemented using Streamlit, allowing users to interactively input features and obtain predictions.

The interface allows users to modify feature values using sliders.

After prediction, the system displays mood classification and popularity.

The recommendation system suggests songs based on similarity.

IV. Experimental Observations

The analysis shows that energy and valence significantly influence mood classification. Songs with high energy and valence are often categorized as energetic or happy, while low values correspond to calm or sad moods.

V. Evaluation

The confusion matrix indicates that the model performs well for distinct classes but may show minor confusion between similar categories.

A. Performance Analysis

The model demonstrates stable performance across different feature combinations. The integration of prediction and recommendation improves overall usability.

B. Error Analysis

Errors mainly occur due to overlapping feature ranges and the limitations of rule-based labeling.

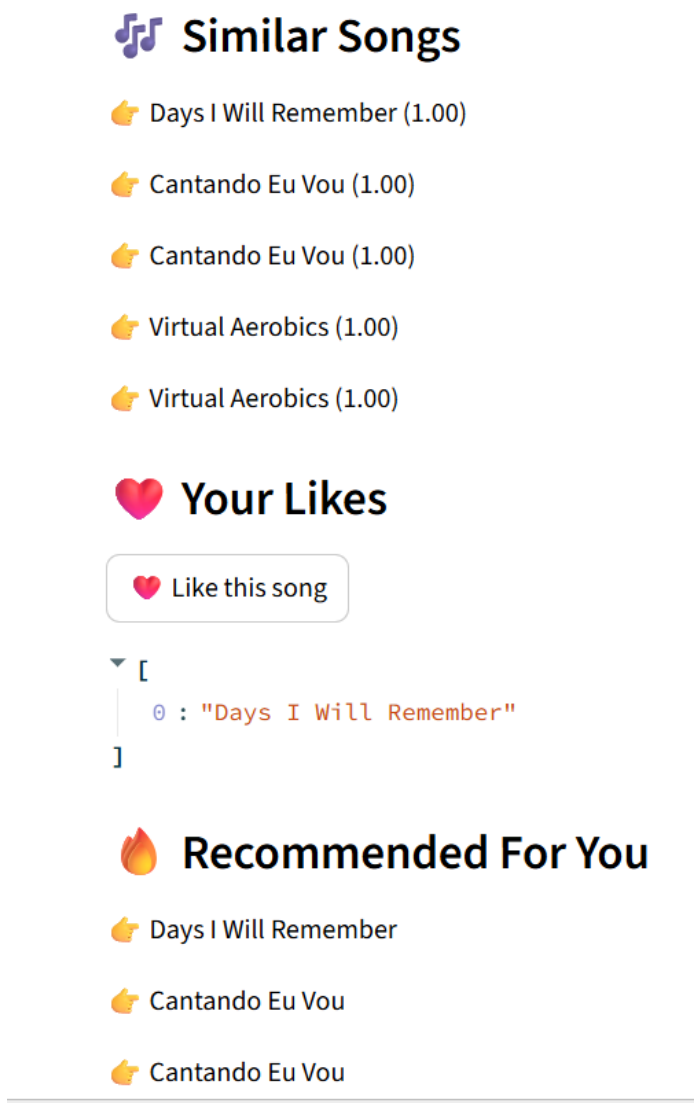

Fig. 3. Recommended Songs Based on Similarity

VI. Extended Analysis

Feature selection plays a critical role in determining model performance. Small variations in feature values can significantly impact predictions, indicating sensitivity in the decision boundaries.

VII. Conclusion

This paper presents a Smart Music Intelligence System that integrates machine learning techniques for mood prediction and recommendation. The system improves user interaction and demonstrates the effectiveness of feature-based modeling.

VIII. Future Work

Future improvements include incorporating deep learning models, real-time user feedback, and additional features such as lyrics and genre.

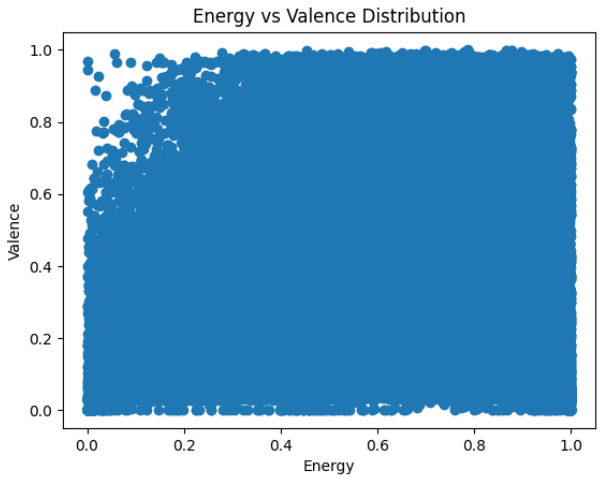

Fig. 4. Relationship Between Energy and Valence Features

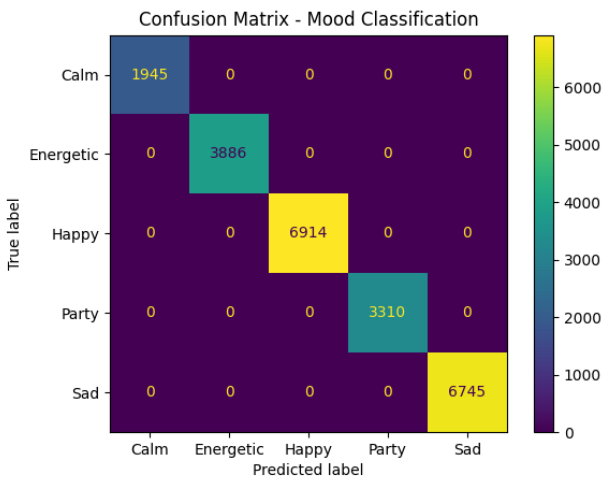

Fig. 5. Confusion Matrix Showing Classification Performance

References

[1] Scikit-learn Documentation
[2] Spotify Dataset
[3] Streamlit Documentation